

Solution Architecture

Date	05 July 2026
Team ID	XXXXXX
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Solution Architecture Diagram

Presentation / Client Layer

Streamlit Web Application
Responsive User Interface



API Gateway / Server

Python • Streamlit Server



Authentication

User Login
Session Management

Core Logic

Document Parsing
Quiz Scoring Engine
Jargon Explainer

External APIs

Google Gemini API



Data / Storage Layer

Browser LocalStorage (Active Session Cache)
Firebase Firestore

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the responsive user interface for uploading text/PDF study materials, taking quizzes, and viewing personalized learning dashboards.	HTML5, CSS, React.js, Streamlit
API Gateway	Routes requests and handles communication between the front-end interface,	Python, Streamlit

	authentication layer, and core AI processing services.	
Core Logic Service	Performs document context parsing, executes interactive quiz grading logic, and runs offline jargon-busting explanations.	Python, FastAPI
Database / Storage	Caches transient session context and offline notes directly in the client browser. Stores secure persistent user credentials, historical quiz scores, and academic roadmaps.	Browser Local Storage, Firebase Firestore / Realtime Database